Acids, Bases and Salts

The big idea

Acids taste sour and turn blue litmus red; **bases** taste bitter, feel soapy and turn red litmus blue. When they react, they cancel each other in **neutralisation**, forming a **salt and water**.

Indicators

Indicator	In acid	In base
Blue litmus	Red	Blue
Red litmus	Red	Blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow
Turmeric	Yellow	Reddish-brown

Olfactory indicators change smell: onion and vanilla lose their smell in a base.

How acids and bases react

KEY REACTIONS

Acid + metal → salt + **hydrogen** (the gas burns with a pop sound)

Acid + metal carbonate/hydrogencarbonate → salt + water + **carbon dioxide** (turns lime water milky)

Acid + base → salt + water (**neutralisation**)

Acid + metal oxide → salt + water (metal oxides are basic)
Base (NaOH) + zinc → sodium zincate + hydrogen

Acids produce **H^+ (H_3O^+) ions** in water; bases (alkalis, if soluble) produce **OH^- ions**. That is why both conduct electricity in water but dry HCl gas doesn't change dry litmus.

DILUTING AN ACID

Always add **acid to water** slowly, with stirring. Adding water to concentrated acid releases so much heat that the mixture can splash out.

The pH scale

pH runs from **0 to 14**. pH 7 is neutral, below 7 acidic, above 7 basic. The lower the pH, the stronger the acid; the higher the pH, the stronger the base. A **universal indicator** shows different colours across the scale.

PH IN EVERYDAY LIFE

Our body works in a narrow range of 7.0–7.8. Tooth decay starts below pH 5.5, so toothpaste (basic) neutralises mouth acid. Antacids (milk of magnesia) neutralise excess stomach acid. Bee stings (methanoic acid) are treated with baking soda. Plants need soil of a suitable pH.

Important salts

Salt	Formula	How it's made / used
Common salt	NaCl	From sea water; raw material for many chemicals
Sodium hydroxide	NaOH	Chlor-alkali process: electrolysis of brine gives NaOH, Cl_2, H_2

Salt	Formula	How it's made / used
Bleaching powder	$CaOCl_2$	Cl_2 + dry slaked lime; bleaching, disinfecting water
Baking soda	$NaHCO_3$	Baking (releases CO_2), antacid, soda-acid fire extinguishers
Washing soda	$Na_2CO_3 \cdot 10H_2O$	Recrystallised sodium carbonate; removes hardness of water, cleaning
Plaster of Paris	$CaSO_4 \cdot \frac{1}{2}H_2O$	Heating gypsum at 373 K; plasters for fractures, toys, decoration

Water of crystallisation: fixed water molecules in a salt crystal. Blue copper sulphate ($CuSO_4 \cdot 5H_2O$) turns white when heated as it loses this water.

Quick recap

- Acids give H^+, bases give OH^-; acid + base → salt + water.
- pH < 7 acidic, 7 neutral, > 7 basic.
- Know the formula, preparation and uses of the six key salts.

Practice questions

1. A solution turns red litmus blue. Its pH is likely to be
   (a) 1 (b) 4 (c) 7 (d) 10
2. The gas released when an acid reacts with a metal is
   (a) carbon dioxide (b) hydrogen (c) oxygen (d) chlorine
3. The chemical formula of Plaster of Paris is
   (a) $CaSO_4 \cdot 2H_2O$ (b) $CaSO_4 \cdot \frac{1}{2}H_2O$ (c) $CaCO_3$ (d) $CaOCl_2$
4. Tooth decay starts when the pH in the mouth falls below
   (a) 7.0 (b) 5.5 (c) 8.5 (d) 3.0
5. Which salt is used to remove permanent hardness of water?
   (a) Baking soda (b) Washing soda (c) Bleaching powder (d) Common salt
6. Phenolphthalein in a basic solution is
   (a) colourless (b) red (c) pink (d) yellow

Answers: 1. d 2. b 3. b 4. b 5. b 6. c